



Post-hoc compression track

Standard fine-tuning (MNRL)

Fine-tuned FP32 baseline
d = 768, float32

INT8 scalar
quantisation

Binary
quantisation

Product
quantisation (PQ)

TurboQuant

Training-time track

Binary-aware fine-tuning (MNRL)

STE BAT

Annealed-tanh BAT

MRL (Matryoshka MNRL)

MRL fine-tune

MRL + STE BAT

MRL + annealed-tanh
BAT

Truncation to d ∈ {32, 64, 128, 256, 512}
+ optional: {Binary, PQ, TurboQuant}

NanoBEIR evaluation pipeline
13 subsets, MRR@10, paired significance tests